

SEMESTER END / BACKLOG SUBJECT EXAMINATIONS – JANUARY 2026

Program	: B.E. – Information Science and Engineering	Semester	: V
Course Name	: Theory of Computation	Max. Marks	: 100
Course Code	: 21 / 22 / 23 / IS54	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

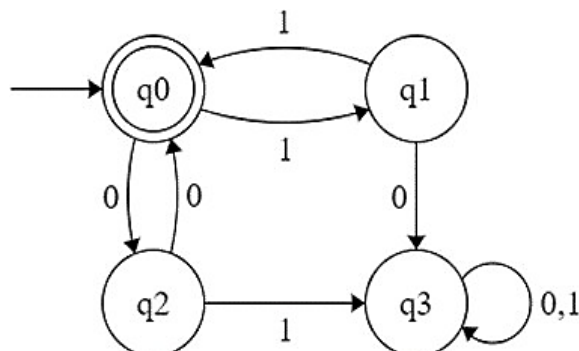
- Discuss the central concepts of automata theory with the help of examples. CO1 (04)
 - Draw a DFA to accept the following language: CO1 (08)
 - $L = \{w(ab+ba) \mid w \in (a,b)^*\}$
 - Strings of a's and b's except those having the substring "aab"
 - Strings of 0's and 1's having three consecutive 0's
 - strings of a's and b's having even numbers of a's and odd numbers b's.
 - With the help of an example, discuss the various components of Deterministic finite automata (DFA). CO1 (08)
- With an example illustrate the process of epsilon closures computation in automata. CO1 (04)
 - Design a DFA for the NFA language $L = \{w \mid w \text{ ends in } 01\}$ using Lazy evaluation. CO1 (08)
 - Differentiate between DFA, NFA and ϵ - NFA. CO1 (08)

UNIT - II

- Construct ϵ -NFA for the regular expression $(a+b)^*ab$. CO2 (05)
 - Minimize the following DFA using table filling algorithm. CO2 (10)

δ	0	1
$\rightarrow A$	B	A
B	A	C
C	D	B
*D	D	A
E	D	F
F	G	E
G	F	G
H	G	D

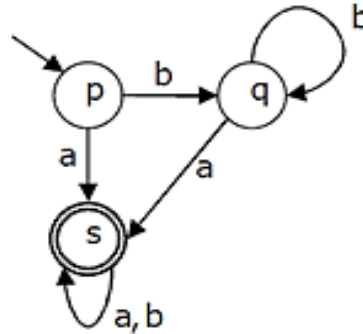
- Convert the following FA to a regular expression using state elimination method-



4. a) What is regular expression? Write regular expressions for the following languages: CO2 (06)
- $L = \{a^n b^m \mid n \geq 4, m \leq 3\}$
 - $L = \{a^m b 2^n \mid m, n \geq 0\}$
- b) Apply $R_{ij}^{(k)}$ method to find the equivalent regular expression for the given finite automata- CO2 (09)

	0	1
$\rightarrow q_0$	q1	q0
$*q_1$	q1	q1

- c) What are indistinguishable states? Are p and q indistinguishable states in the given FA for the input string 'bba'? Justify your answer. CO2 (05)



UNIT - III

5. a) Convert the given grammar into Chomsky normal form by eliminating ϵ production, unit production and useless production. CO3 (10)
- $S \rightarrow AaBa \mid ASBb$
 $A \rightarrow aAbB \mid a \mid B$
 $B \rightarrow SbS \mid AC \mid bb \mid \epsilon$
 $C \rightarrow aC \mid c$
 $D \rightarrow d$
- b) With the help of an example, explain the process of constructing a parse tree. CO3 (05)
- c) Is the following grammar ambiguous? CO3 (05)
- $S \rightarrow aB \mid bA$
 $A \rightarrow aS \mid bAA \mid a$
 $B \rightarrow bS \mid aBB \mid b$

6. a) Design context free grammars for the following language. CO3 (10)
- $L = \{w \mid n_a(w) = n_b(w), w \in (a+b)^*\}$
 - $L = \{a^n b^m \mid n \geq 0, m > n\}$
 - $L = \{w : n_a(w) \bmod 2 = 0 \text{ where } w \in (a,b)^*\}$
 - $L = \{a^n b^{2n} : n \geq 0\}$
 - $L = \{ww^r \text{ where } w \in \{a,b\}^*\}$
- b) Illustrate the process of removing useless production from a grammar. CO3 (05)
- c) Discuss leftmost derivation and rightmost derivation with the help of an example. CO3 (05)

UNIT- IV

7. a) Design a push-down automata to accept the language $L = \{a^n b^{3n}, n \geq 1\}$. Draw the transition diagram. With the sequence of IDs show that the input string aabbbbbbb is valid. CO4 (10)
- b) Explain the following set notation. What language does it indicate? CO4 (05)
- $L = \{w \mid (q_0, w, Z_0) \vdash^* (q, \epsilon, \epsilon)\}$

- c) Are the following transitions for NPDA? Give justification for your answer. CO4 (05)
- i) $\delta(q_0, (, Z_0) = (q_1, Z_0)$
 - ii) $\delta(q_0, [, Z_0) = (q_1, [Z_0)$
 - iii) $\delta(q_1, (, () = (q_1, (()$
 - iv) $\delta(q_1, (, [) = (q_1, ([)$
 - v) $\delta(q_1, [, () = (q_1, [()$
 - vi) $\delta(q_1, [, [) = (q_1, [[)$
 - vii) $\delta(q_1,), () = (q_1, \epsilon)$
 - viii) $\delta(q_1,], [) = (q_1, \epsilon)$
 - ix) $\delta(q_1, \epsilon, Z_0) = (q_0, Z_0)$
8. a) What type of strings are accepted by the following PDA? Describe each of the transitions. CO4 (06)
- i) $\delta(q_0, a, Z_0) = (q_0, aZ_0)$
 - ii) $\delta(q_0, a, a) = (q_0, aa)$
 - iii) $\delta(q_0, b, a) = (q_1, \epsilon)$
 - iv) $\delta(q_1, b, a) = (q_1, \epsilon)$
 - v) $\delta(q_1, \epsilon, Z_0) = (q_f, Z_0)$
- b) Design a push-down automata to accept the language $L = \{N_a(w) = N_b(w) \mid w \in (a+b)^*\}$. Is it DPDA? Justify your answer. CO4 (08)
- c) Construct a PDA for the following CFG- CO4 (06)
- $S \rightarrow aA B C, A \rightarrow a B \mid a, B \rightarrow b A \mid b, C \rightarrow a.$
- UNIT - V**
9. a) Prove that every language accepted by a Multitape TM is recursively enumerable. CO5 (10)
- b) With the help of example, explain NP-Hard and NP-complete. CO5 (05)
- c) Illustrate storage in the state in turing machine. CO5 (05)
10. a) Design a Turing machine to accept the language $L = \{0^n 1^n \mid n \geq 1\}$. Prove that string the "000111" is valid or not. CO5 (10)
- b) Illustrate the working of a turning machine using block diagram. CO5 (05)
- c) Explain the halting problem in a turning machine. CO5 (05)
